Natural-Language Controls for Short-Video Recommenders

An offline study of controllability

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Experiment Two results

100 fresh synthetic requests with model-generated references.
Llama: 76 exact, 23 rejected, one accepted mismatch.
Rules: 11/100 exact matches.

Intended-control success combines requested counts
with a change in ranked category exposure.
Llama: 20.145%; item feedback: 3.815%.
Conditional requests: no control successes.
NDCG@10: 0.02580 Llama versus 0.03184 item feedback.

The earlier exposed practice score of 73/100 is separate. Model-generated references are not human validation.

Research question

Can one natural-language interaction satisfy a richer intended change than one item reaction?

- A compound request can specify several categories, strength and exceptions.
- The LLM must also outperform a rule parser to establish its own contribution.
- Offline compliance and historical prediction measure different outcomes.

The claim that language *feels more natural* requires human evidence beyond this study.

Prior work supports a control question

Source	Relevant evidence
Mozilla, 2022	Dislike prevented 12% of unwanted recommendations and Not interested 11%, relative to no feedback.
Ramos et al., 2024	Positive profile edits raised target Coverage@10 without further fine-tuning. Amazon-MT RMSE: 0.941 versus MF 0.925, where lower is better.
Sanner et al., 2023	Language preferences were competitive in a near cold-start movie setting.
Chen and Pu, 2012	Critiquing systems already studied richer explicit preference feedback.

Historical results in different settings. Bibliographic details are in paper/references.bib.

Translation and enforcement are separate

Stage	Role
Recommender	Frozen SASRec-style scores for the candidate catalogue.
Translator	Llama 3.1 8B or rules compile language into a validated command.
Enforcer	Shared deterministic reranking applies the command.
Explanation	Verified facts describe the displayed item and control.

A correct hard exclusion comes from the enforcement code. It does not demonstrate language understanding.

Data and information boundaries

- KuaiRec cohort: **865 users**, **2,593 candidates**, 31 opaque category identifiers.
- Positive event: watch ratio > 2, an engagement proxy.
- Global chronological windows: nominal 70% training, 15% request history, 15% held-out evaluation.
- Requests use training and request history only.
- Quality target: first eligible held-out positive item.

Synthetic requests and numeric category labels limit claims about real preferences and language.

Filtered cohort counts, not the upstream KuaiRec matrix size. The candidate list could include items users had already seen.

Five matched configurations

ID	Mechanism
A	One simulated reaction changes one historical exemplar item.
B	A plus display-only explanations.
C	Free text compiles into a category control.
D	An editable preference sentence compiles into a category control.
E	Text, profile and explanations, with duplicate commands removed.

One boost and one mute request per user, under seeds 42, 43 and 44. The item-feedback proxy is an explicit design choice.

A is not a reconstruction of a commercial feedback algorithm. B must rank identically to A.

Parser results on synthetic language

Language material	Llama correct	Meaning
Templated inputs	120/120	Canonical task
Broader synthetic set	79/100	Below 85% target
Adversarial subset	2/10	Major weakness

- 79% exceeds the 70% floor. It misses the 85% target.
- The rule parser scores 81/100 on the same broader set.
- The 120 canonical wording patterns informed development.

The adversarial subset belongs to the 100 cases. These finite synthetic sets do not establish general language reliability.

Experiment 1: category enforcement

Top-10 exposure	A	C, D and E
Boost target share, TCP	11.6%	79.2%
Muted target share, TCER	6.1%	0.0%

- Hard exclusion guarantees zero exposure when parsed correctly.
- Category-wide boosts affect more items than A's single-item adjustment.
- Rule and LLM paths produce identical feeds for these canonical Experiment 1 requests.

Averages over 865 users and three seeds. This is enforcement evidence, not an LLM advantage.

Steering has a predictive quality cost

Boost requests	A	C
NDCG@10	0.03012	0.01637

45.7% relative decrease.

Paired mean change: -0.01375

95% user-bootstrap interval: $[-0.02086, -0.00655]$.

This post-hoc secondary contrast measures agreement with a historical next item. It does not measure user dissatisfaction after steering.

Post-hoc exploratory interval, outside the frozen primary BH family.

A weak backbone limits the quality claim

Uncontrolled ranking	HR@1
SASRec, seed 42	0.000%
SASRec, seed 43	0.347%
SASRec, seed 44	0.462%
Recency popularity	7.977%

Recency pools all users' request history. The model's weights use the older training window. This comparison is descriptive. The absolute 10 percentage point non-inferiority margin is uninformative at this baseline.

Post-hoc uncontrolled baselines. A pooled HR@1 interval includes zero. No accuracy gain is established.

Experiment 1 conclusion

1. Deterministic category control works: TCP 11.6% to 79.2%, TCER 6.1% to zero.
2. Llama parsed 120/120 templated inputs, 79/100 broader synthetic inputs and 2/10 adversarial cases.
3. All 10/10 explanation fact selections equal the template.
4. Boost NDCG@10 falls 45.7%. Non-inferiority is uninformative.
5. Backbone HR@1 is 0.0–0.46%, below recency popularity's 8.0%.

The study establishes bounded enforcement and its limitations. It does not establish an LLM advantage or useful quality preservation.

What the fresh experiment tests

- Control success versus rules and item feedback.
- Requested exposure movement versus item feedback.
- Historical next-item accuracy as a separate outcome.
- Earlier practice results were used during development and are separate from the fresh test.
- Synthetic requests do not measure human experience.

These claims concern this fixed system and model-generated references.

Four classes of intent

Class	Illustrative request	Top-10 requirement
Compound	More X and less Y	$n(X) \geq 2$, $n(Y) \leq 1$
Graded	A bit more X	$2 \leq n(X) \leq 4$
	Much less X / none of X	$n(X) \leq 1$ / $n(X) = 0$
Floor	Less X, but not zero	$1 \leq n(X) \leq 2$
Conditional	More X only if also Y	$n(X \cap Y) \geq 2$ and no increase in $X \setminus Y$ count

Success also requires a full list and positive intended movement. Bands are operational conventions.

X and Y denote opaque categories. Conditional means the same item carries both labels. These are examples, not test requests.

A matched interaction budget

- A gets one positive or negative reaction on one historical exemplar.
- Rule and LLM get the complete utterance once and share the same ranker.
- Assignments use only earlier history and a committed SHA-256 scheme.
- Channels share permitted pre-test information. Recency and SASRec use that information differently.

The budget matches interaction count. It does not equalize semantic content, user effort or every possible feedback algorithm.

Prospective design

The evaluation procedure was fixed before the 100 new requests were created.

100 balanced requests across four classes.
Two separate model annotation passes.
Reference labels saved before comparison.

Agreement measures model consistency, not human validity.

Both passes used the same reported model family.

Fresh requests and reference labels

100 selected requests, 25 in each of four classes.
31 target category IDs covered by the fixed plan.

Screened against 100 exposed and 48 development texts.
No duplicate or frozen close-paraphrase flags selected.

Two passes with the same model family agreed
on all 100 initial commands.
No human adjudication was performed.

Lexical screening does not prove semantic or pretraining novelty. Model agreement is not human ground truth.

Earlier development results

Validated model-reference agreement: Llama 73/100; rules 18/100.
All 24 Llama validation errors remain counted.

Induced success: Llama 23.81%; rules 8.40%;
item feedback 3.92%.
NDCG@10: Llama 0.02461; item feedback 0.03128.

41,508 ranking records were checked.

These earlier questions had been seen during development and do not supply an unseen test.

Prespecified analysis

User is the analysis unit. Average three seeds within each request/class, then equal eligible classes per user.

5,000 paired user bootstrap samples, seed 2026.
Pratt one-sided Wilcoxon with joint BH over 11 tests.

Keep failures, infeasibility and exclusions visible.
Report control, HR@1 and NDCG@10 together.

Analysis choices were fixed before reviewing results and were not tuned to them.

Fresh model-reference evaluation results

Validated parser agreement: Llama 76/100; rules 11/100.
Llama operational/validation failures: 23, retained.

Induced success: Llama 20.145%; rules 3.854%;
item feedback 3.815%. Conditional requests: 0 successes.
NDCG@10: Llama 0.02580; item feedback 0.03184.

41,520 ranking rows; 2,595 full score vectors.

These measured control and quality outcomes concern model-authored synthetic requests, not independent human validation.

Contribution and limits

Under model-generated reference labels, language control exceeded the rule parser and single-item feedback on the prespecified control measures.

Translation, enforcement and quality remain observable. Failures remained in every reported comparison.

Human experience, naturalistic language and other LLM models require separate evidence.

The weak backbone, opaque categories and historical next-item proxy limit recommendation-quality interpretation.

Interpretation of the results

- Language translated richer multi-category requests.
- Deterministic rules applied validated controls.
- Pooled control gains came with lower next-item NDCG@10.
- Conditional requests had no successful controls.
- Synthetic wording and model references limit generalization.

Category control and historical next-item prediction are separate outcomes.

What these results can establish

- The 100 synthetic requests were evaluated across 865 users.
- Reference commands came from two passes using the same model family.
- Human understanding and satisfaction were not measured.
- Full E1 replay on Windows and Colab remains unverified.

The separate research archive retains the full study records.

Student response check, 20 October

Vibhor reports completing the original 100-question worksheet himself without seeing reference answers.

Under the study's stated annotation rules, his responses matched 99 of 100 initial AI-authored labels.

This is a descriptive student-versus-AI comparison, not agreement between independent human annotators.

The original human-reference evaluation still needs adjudication and separate parser and ranking analysis.

Backup: metric definitions

TCP@10 Target-category items divided by ten for boost requests.

TCER@10 Target-category items divided by ten for mute requests.

HR@1 One if the first item equals the held-out target, otherwise zero.

NDCG@10 For one relevant item at rank $r \leq 10$, $1/\log_2(r+1)$. Zero if absent.

Both exposure metrics have the same arithmetic and opposite desired directions. Historical quality does not directly measure preference satisfaction after intervention.

Backup: backbone and control settings

Backbone	Compact SASRec-style implementation
Hidden size / context	32 / 50 items
Attention	Two blocks, one head
Training	20 epochs, Adam, learning rate 0.001
Batch / dropout	64 / 0.2
Training seeds	42, 43, 44
Experiment 1 control	Normalized score +0.25, or hard mute
Experiment 2 strengths	0.10, 0.25, 0.50
Language backend	Llama 3.1 8B, temperature 0

The backbone is independently implemented. The study does not claim an exact SASRec paper reproduction.

Backup: why non-inferiority is uninformative

Pooled Experiment 1 quantity	Value
A HR@1 baseline	0.2505%
C minus A, mean	$+0.193$ pp
Paired 95% interval	$[-0.058, +0.501]$ pp
Simultaneous lower bound	-0.07707 pp
Absolute proposed threshold	-10 pp
Relative 10% loss threshold	-0.02505 pp

The absolute reading numerically passes while allowing the entire small baseline to disappear. The relative reading fails. Only eight of 865 users have nonzero paired HR@1 differences.

Both operations pooled, distinct from the boost-only NDCG contrast. pp means percentage points.

Backup: intended-direction movement

$$D(S) = \frac{\sum_{r=1}^{10} \mathbf{1}[i_r \in S] / \log_2(r+1)}{\sum_{r=1}^{10} 1 / \log_2(r+1)}$$

ΔD is the channel value minus its uncontrolled value.

Class	Signed movement M
Compound	$\min(\Delta D(X), -\Delta D(Y))$
Boost / reduction	$\Delta D(X)$ / $-\Delta D(X)$
Floor	$-\Delta D(X)$, zero if the floor is violated
Conditional	$\Delta D(X \cap Y) - \max(0, \Delta D(X \setminus Y))$

Induced satisfaction requires a full list, the class band and $M > 10^{-12}$. This operational measure is not Shannon information.

Backup: study records

100 requests and 200 retained parser responses.
3,460 assigned user/request pairs.
41,520 ranking rows; 2,595 score vectors.

The study saved its prompts, model settings, inputs and calculations before the result comparison.

Full E1 replay on Windows and Colab remains open.

Backup: selected references

- Ricks and McCrosky (Mozilla), 2022. *Does this button work? Investigating YouTube's ineffective user controls.*
- Ramos et al., 2024. *Transparent and Scrutable Recommendations Using Natural Language User Profiles*. arXiv:2402.05810.
- Sanner et al., RecSys 2023. *Large Language Models are Competitive Near Cold-start Recommenders for Language- and Item-based Preferences*. arXiv:2307.14225.
- Chen and Pu, 2012. *Critiquing-based recommenders: survey and emerging trends*. UMUAI 22, 125–150.
- Kang and McAuley, ICDM 2018. *Self-Attentive Sequential Recommendation*.
- Gao et al., CIKM 2022. *KuaiRec: A Fully-observed Dataset and Insights for Evaluating Recommender Systems*.

Complete bibliographic records and source links: paper/references.bib.